

GRADE 7 · READING INFORMATIONAL TEXT · RI.7.1

Evidence in Informational Text

Cite several pieces of textual evidence to support analysis of explicit statements and inferences drawn from an informational text about coral reef decline.

Lesson Map · Passage · Guided · Partner · Independent Practice · Small Group · Answer Key

What You Will Learn Today

Learning Goals

Understand RI.7.1: Cite several pieces of textual evidence to support analysis of explicit...

Read an original passage closely

Support every answer with text evidence

Work through guided, partner, and independent practice

How the Lesson Runs

Warm-up and passage

Guided practice together

Partner discussion

Independent + small-group work

Exit ticket and answer key

Unpacking the Standard

In informational text, explicit evidence is stated directly (a fact, a statistic, a direct claim). An inference is a reasonable conclusion built from evidence plus background knowledge. Strong analysis of informational text, like literary text, usually needs more than one piece of evidence.

Modeled Example

The article states directly: "Reef temperatures rose 1.5 degrees in a decade." Is that explicit or an inference?

Explicit — it is a specific, stated fact with a number.

The article notes that fish populations near the reef dropped by half in the same period. What might you infer, combining both facts?

You could infer that warming water conditions are connected to the decline in fish populations, since both changes happened in the same area and timeframe.

The Quiet Bleaching

1. Coral reefs cover less than one percent of the ocean floor, yet they support roughly a quarter of all known marine species. Off the coast of a small research station, marine biologist Dr. Amara Osei has tracked one reef system for eleven years.
2. In her first year, the reef's coral cover measured at 68 percent, glowing with the greens and purples typical of a healthy system. By her ninth year, cover had dropped to 31 percent. Much of the remaining coral had turned a pale, bone white.
3. This whitening, called bleaching, happens when coral, stressed by unusually warm water, expels the colorful algae living in its tissue. The algae provide the coral with most of its food through photosynthesis. Without them, coral can survive for a period, but not indefinitely.
4. Dr. Osei's temperature logs show the surrounding water has warmed by 1.5 degrees Celsius over the eleven-year study, a change that sounds small but crosses a threshold that many coral species cannot tolerate for extended periods.
5. Fish surveys from the same reef tell a related story. In year one, divers counted an average of 340 fish per survey transect. By year nine, that number had fallen to roughly 170 — half of the original count.

The Quiet Bleaching, continued

7. Dr. Osei now spends part of each season mapping these cooler refuges. "If warming continues," she wrote in her most recent field report, "these pockets may be what the rest of the reef eventually has to rebuild from — if it rebuilds at all."

Warm-up · Before You Read

1. What explicit statistic does paragraph 1 give about coral reefs' share of the ocean floor?
2. What explicit percentage does paragraph 2 give for coral cover in year one versus year nine?

Guided Practice

3. What can you infer about the health of the reef system from that drop in coral cover? Cite the evidence.

4. What does paragraph 3 state explicitly about why coral bleaches?

Guided Practice

5. What can you infer about why bleached coral is at risk, using evidence from paragraph 3?

6. What explicit temperature change does paragraph 4 report?

Partner Discussion

7. What explicit numbers does paragraph 5 give for the fish survey counts?

8. What can you infer about the relationship between coral decline and fish decline? Cite evidence from at least two paragraphs.

Partner Discussion

9. What explicit detail does paragraph 6 give about the northern section of the reef?

Independent Practice

10. What can you infer about why the northern section has fared better? Cite evidence.

11. What does Dr. Osei state explicitly in her field report, quoted in paragraph 7?

Independent Practice

12. What can you infer about Dr. Osei's level of concern from her quoted statement? Cite evidence.

13. Cite two pieces of explicit evidence that show the reef has changed over the eleven-year study.

Independent Practice

14. Cite evidence that supports the inference that not all parts of the reef are affected equally.

15. Why might Dr. Osei choose to study the cooler northern section more closely, based on the text?

Small-Group Work

16. Is it explicit or an inference that warmer water is the main cause of the coral decline? Explain using the text.

17. Combine two pieces of evidence to support the inference that the reef's decline is an ongoing trend, not a one-time event.

Small-Group Work

18. What inference can you draw about the future of the reef if water temperatures continue to rise? Cite evidence.

19. Explain why a scientist like Dr. Osei tracks a reef for eleven years rather than studying it once.

Exit Ticket

20. Write one full inference about the reef's future, with two pieces of cited evidence supporting it.

Answer Key (1 of 4)

1. Coral reefs cover less than one percent of the ocean floor.
2. 68 percent in year one, dropping to 31 percent by year nine.
3. The reef's health has significantly declined; a drop from 68 percent to 31 percent coral cover over eight years suggests a serious, ongoing loss rather than a temporary change.
4. Coral expels the colorful algae living in its tissue when stressed by unusually warm water.
5. Bleached coral is at risk of starving, since paragraph 3 states the algae "provide the coral with most of its food" and that without them coral "can survive for a period, but not indefinitely."

Answer Key (2 of 4)

6. The water has warmed by 1.5 degrees Celsius over the eleven-year study.
7. 340 fish per transect in year one, dropping to roughly 170 by year nine.
8. The two declines appear connected; paragraph 2 shows coral cover fell from 68 to 31 percent, and paragraph 5 shows fish counts fell by half over a similar period, suggesting the fish rely on the reef habitat.
9. It has retained nearly 60 percent coral cover throughout the study, in a shaded, deeper section where water stays several degrees cooler.
10. Cooler water appears to protect coral from bleaching; paragraph 6 states this section stays "several degrees cooler" and has retained far more coral cover than the reef overall.

Answer Key (3 of 4)

11. "If warming continues, these pockets may be what the rest of the reef eventually has to rebuild from — if it rebuilds at all."

12. She seems seriously concerned about the reef's long-term survival; the phrase "if it rebuilds at all" suggests she is uncertain whether recovery is possible.

13. The coral cover dropped from 68 to 31 percent (paragraph 2), and the water temperature rose by 1.5 degrees Celsius (paragraph 4).

14. Paragraph 6 states the northern section has retained nearly 60 percent coral cover, much higher than the reef's overall 31 percent average.

15. Because it may hold clues for how the reef could recover, since paragraph 7 suggests these cooler pockets might be what the rest of the reef "has to rebuild from."

Answer Key (4 of 4)

16. It is a supported inference rather than a single explicit statement; paragraphs 3 and 4 explain the bleaching process and rising temperature, and paragraph 6's cooler-water comparison supports the connection, though no single sentence states "warm water is the main cause."

17. Coral cover fell steadily from 68 percent (year one) to 31 percent (year nine), and fish counts fell in a similar pattern across the same nine years, showing a consistent decline over time rather than a single event.

18. The reef may continue declining and could struggle to recover; Dr. Osei's own statement in paragraph 7 raises doubt about whether the reef "rebuilds at all" if warming continues.

19. Tracking it over many years allows her to see patterns and trends (like the steady drop in coral cover and fish counts) that a single visit could not reveal, and to identify areas like the cooler northern section that resist decline.

20. Answers vary; require a stated inference plus two specific, correctly cited details from the passage that reasonably support it.

Exit Ticket

Answer on a sticky note

Answer the exit-ticket question from today's lesson.

Cite at least one piece of evidence from the passage.

Remember

Name your claim or answer

Quote or cite the evidence

Explain how it supports your answer